HYDRAULIC SYSTEM TROUBLESHOOTING GUIDE

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1. Safety Warning

WARNING: Hydraulic fluid under high pressure can cause serious injury or death. Never work on a pressurized system. Always perform Lockout-Tagout (LOTO), release residual pressure, and verify zero energy state before opening any hydraulic line or component.

2. Common Problems and Corrective Actions

2.1 Pump Noise / Cavitation

- Check suction strainer for blockage or collapse.
- Verify oil level in the reservoir is above the minimum mark.
- Confirm oil viscosity is correct (ISO VG 46 is recommended for most industrial systems).
- Inspect suction line for air leaks at fittings and seals.

2.2 System Overheating

- Check cooler efficiency and cooling water/air flow.
- Verify relief valve setting is not too close to normal working pressure.
- Look for internal leakage in cylinders, valves, or the pump itself.
- Ensure the correct oil grade is being used for the ambient temperature.

2.3 Slow Actuator Movement

- Measure actual pump flow rate and compare with nameplate value.
- Check directional control valve for sticking or incomplete shifting.
- Inspect for worn cylinder seals causing internal bypass.
- Confirm that the load is within the design capacity of the system.

2.4 Pressure Not Building Up

- Check if the relief valve is stuck open or set too low.
- Inspect pump for worn parts (vane, gear, or piston wear).
- Look for major external leakage or burst hose.
- Verify that the compensator on a variable displacement pump is functioning.

3. Recommended Oil Cleanliness

Maintain oil cleanliness according to ISO 4406. Target cleanliness code for most industrial hydraulic systems is 18/16/13 or better. Change filters when the differential pressure indicator shows blockage. Take oil samples every 6 months for analysis.

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